

03 | OVERVIEW



Exploring Localized Climate Health: A Case Study of Crab Hill, Barbados

Abstract

Crab Hill is a quiet farming community of about 662 people at the northern tip of Barbados. Researchers worked with residents to understand how flooding, invasive seaweed, poor roads, and unreliable water affect daily life and health. This lesson shows how a local case study can reveal the ways climate change and infrastructure combine to shape a community's well-being.

Introduction

Crab Hill lies in the parish of St. Lucy, about 30 kilometers northeast of Bridgetown. It was once an Arawak settlement and later a plantation; its name comes from the many crabs that came ashore in the rainy season. Today it is the most populated settlement in St. Lucy, with strong churches, a cricket ground maintained by 100 club members, and small local businesses. Like much of Barbados, Crab Hill has not been hit directly by a major hurricane in decades. But climate change is increasing flooding and warming the ocean. Through the Advancing Community Climate Resilience Project, researchers from Northeastern University and the University of the West Indies asked residents what helps them cope and what puts their health and livelihoods at risk.

How Do Scientists Study This?

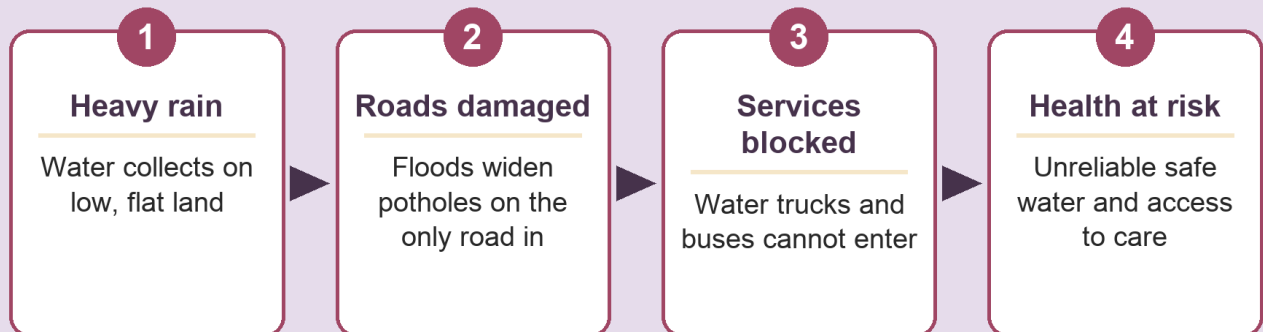


FROM THE ARTICLE

The northern coast of Barbados near Crab Hill.
Image: ACCRP StoryMap, Global Resilience Institute.

In late July 2024, the research team held co-development sessions with Crab Hill residents using the Integrated Resilience Enhancement System (I-RES). Residents identified community assets in four domains: physical, economic, governance, and social. They also pointed out places where floodwater collects, such as Spencer's Bottom and the North Stars Cricket Ground. Analysts combined these observations with the 2021 Barbados census, national climate reports, and records of past storms. They built a GIS flood-risk map from digital elevation, land use, and distance to waterways. Together, the local knowledge and the data showed how hazards, infrastructure, and health are connected in one small place.

How flooding reaches residents' health



Study the Process

Identify assets

Residents named strengths such as churches, the cricket ground, local shops, and the district emergency organization.

Locate hazards

Community members and GIS maps showed where flooding collects and where sargassum washes ashore.

Trace health effects

Researchers linked roads, water supply, and seaweed to residents' health and access to services.

What Did They Discover?

Crab Hill's low, flat coastal land makes it highly susceptible to flooding, and Barbados experiences a major flood event about every twelve years. There is only one road into the community, and most roads are dirt tracks in poor condition. Flooding worsens potholes, so water delivery trucks and public buses often cannot get in. Residents reported poor drinking-water quality; the government linked this to aging pipelines and silt at a pumping station, a problem climate change is likely to accelerate. Sargassum seaweed, which has increased with warming oceans and fertilizer runoff, releases sulfur as it rots. Residents with respiratory conditions can react to the smell, and fewer visitors come to the beach.

Crab Hill at a glance

662

residents, the largest settlement in St. Lucy (2021)

1

road into the community, mostly unpaved

12 yrs

typical gap between major floods in Barbados

Churches act as a linchpin during crises, mobilizing help and reaching vulnerable residents. In St. Lucy, 74% of people identify as Christian.

03 | TAKEAWAYS

Takeaways

Health is local

Roads, water pipes, and beaches all shape residents' health when climate hazards strike.

Hazards connect

Flooding damages roads, which then blocks water trucks and buses from reaching the community.

Faith builds resilience

Churches and close social ties help residents respond and support one another in a crisis.

Key Terms

Sargassum

A type of floating seaweed that has become overly abundant due to warmer oceans and fertilizer runoff. It releases sulfur gas as it decomposes.

Siltation

The build-up of fine soil particles in water systems, which can clog equipment and reduce water quality.

Infrastructure

The basic systems a community depends on, such as roads, water pipes, power lines, and drainage.

Community assets

Existing or potential resources, such as buildings, organizations, and relationships, that can help a community reach its goals.

Climate health

The ways changes in climate and environment affect human health, from air and water quality to access to care.

03 | ACTIVITY

Activity

Connect the Chain

Students build cause-and-effect chains that link climate hazards in Crab Hill to residents' health.

Materials: Index cards; String or yarn; Tape

1. Write each of these on a card: warming ocean, sargassum, sulfur smell, flooding, potholes, water trucks, drinking water, respiratory symptoms.
2. Arrange the cards into chains that show how a climate hazard leads to a health effect. Connect them with string.
3. Add one card showing a community asset, such as churches or the DEO, that could interrupt a chain.
4. Compare chains with another group. Did anyone find a connection you missed?
5. Write two sentences explaining which chain you would break first and why.

References

Global Resilience Institute, Northeastern University, and University of the West Indies. Advancing Community Climate Resilience Project: Crab Hill, Barbados (ArcGIS StoryMap).
<https://storymaps.arcgis.com/stories/d0081be96ae64e3d8a01f27e9c5214f9>